

DOUBLE JEOPARDY: The Vulnerability Spiral of Compound Climate Risk Across 5 Islands — Figures

Manuscript: DOUBLE JEOPARDY: The Vulnerability Spiral of Compound Climate Risk Across 5 Islands

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Figures, in order

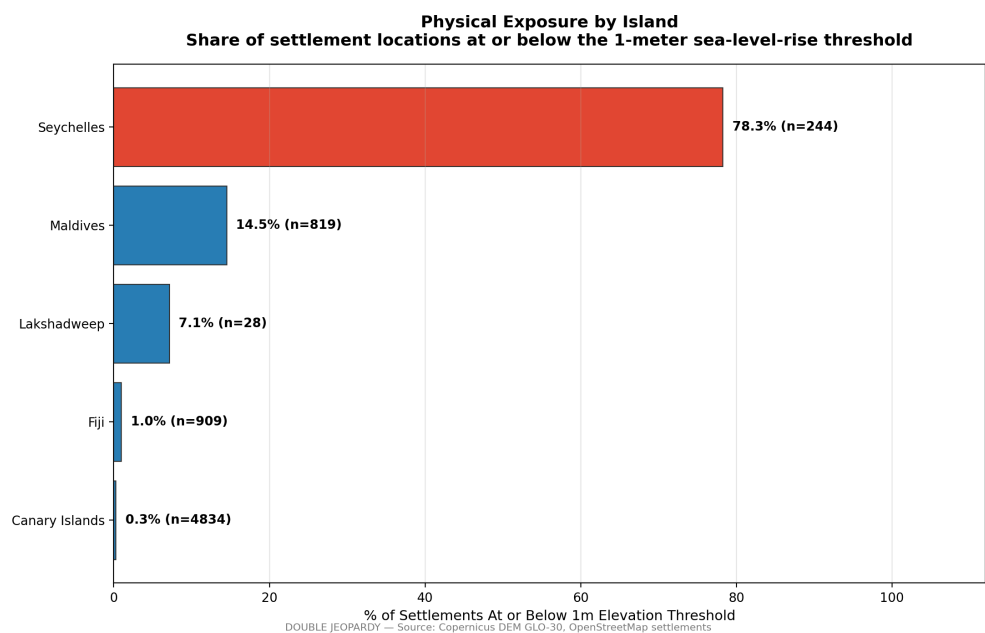


Figure 1. Physical exposure by island, measured as the share of settlement locations at or below the 1 meter sea level rise threshold. Seychelles shows by far the highest settlement based exposure in this sample; Maldives and Lakshadweep, both low lying coral atolls, sit in a moderate middle band; Fiji and the Canary Islands, both volcanic and mountainous, remain lowest.

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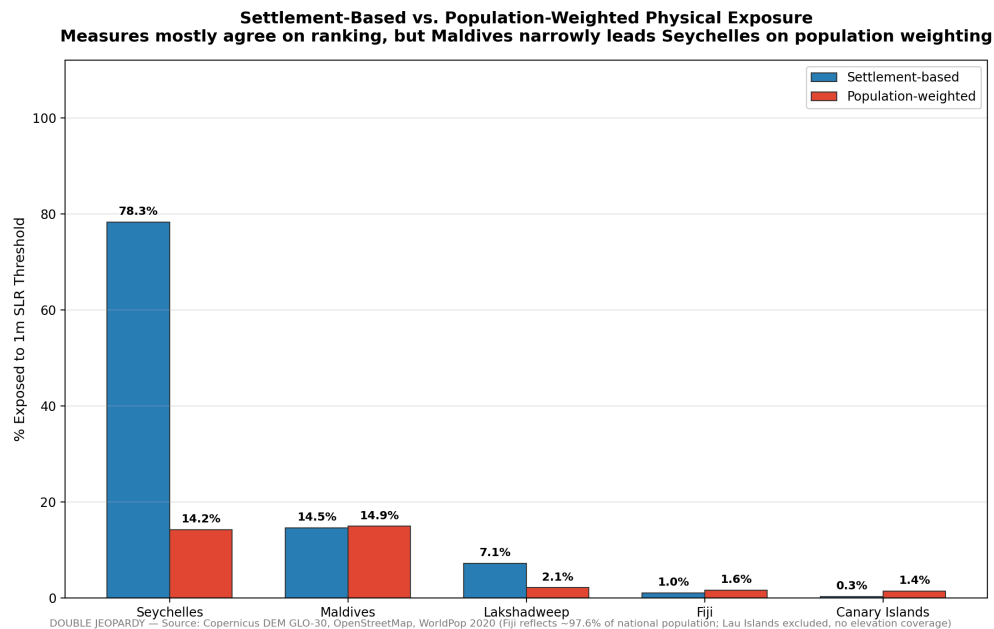


Figure 2. Settlement based versus population weighted physical exposure by island. The 2 measures mostly agree on island ranking (Canary Islands lowest throughout), but they diverge at the top, where Seychelles remains highest on settlement based exposure while Maldives now narrowly leads on population weighted exposure after a boundary polygon correction (see Section 4.4 of the manuscript). An earlier version of this analysis had shown Lakshadweep as the highest population weighted exposure island; that turned out to be an artifact of a since corrected Digital Elevation Model (DEM) coverage gap, and it wasn't a real finding at all. The 2 measures still diverge sharply in magnitude for Seychelles.

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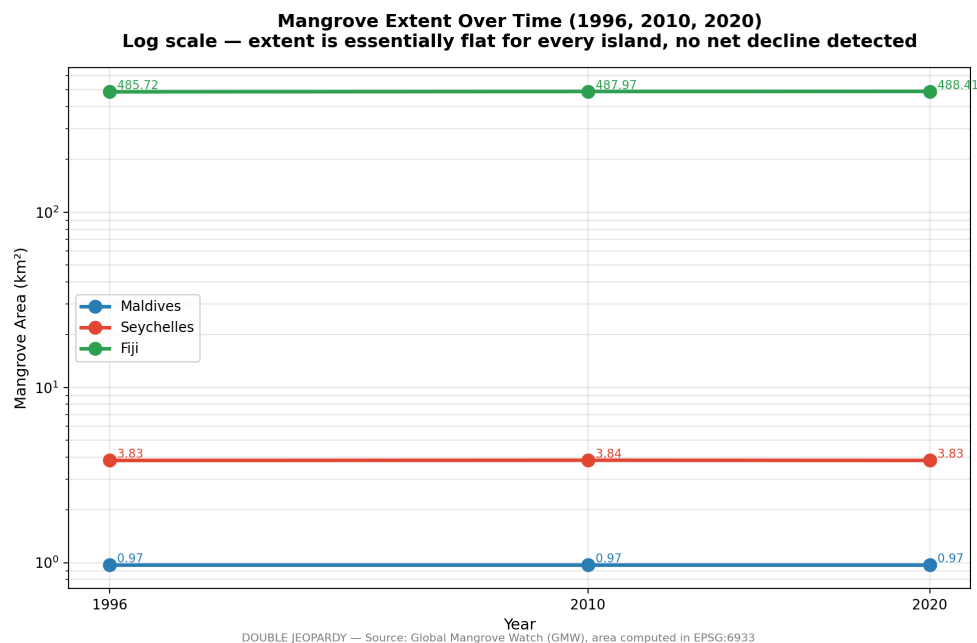


Figure 3. Mangrove extent (square kilometers, log scale) at 3 independent time points (1996, 2010, 2020) for the 3 islands where mangroves are present. Extent remains essentially flat for all 3 islands, with no measurable decline detected.

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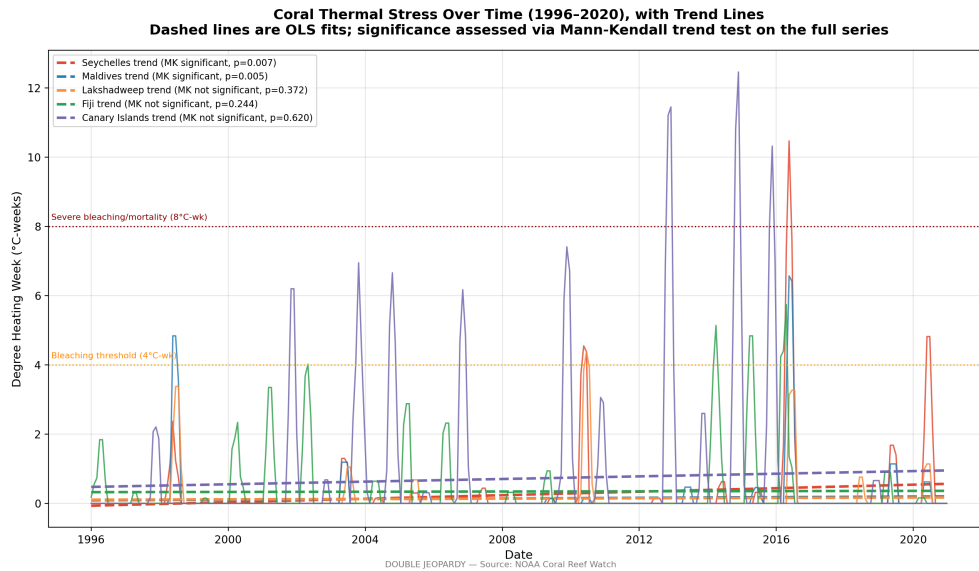


Figure 4. Coral thermal stress (Degree Heating Week) over the full 1996–2020 record for all 5 islands, with ordinary least squares (OLS) trend lines. The seasonal Mann-Kendall trend test finds a statistically significant increasing trend for Maldives ($p=0.0046$) and Seychelles ($p=0.0069$) only.

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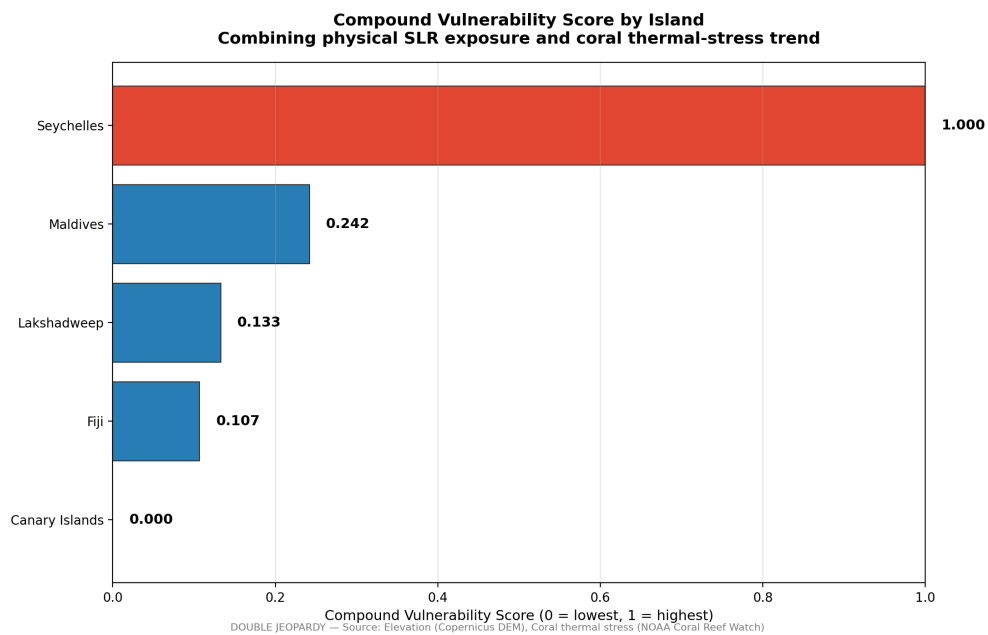


Figure 5. Compound Vulnerability Score across the 5 study islands, combining normalized physical sea level rise exposure and long term coral thermal stress trend using equal weighting. Seychelles is the most vulnerable island on the settlement based exposure measure and on the compound score; population weighted exposure alone now shows Maldives narrowly ahead instead (see Section 3.1 of the manuscript). The compound score's contribution here is confirming that the settlement based ranking holds up once each island's coral thermal stress trend gets factored in on top of exposure. There was no disagreement between indicators here that needed resolving.

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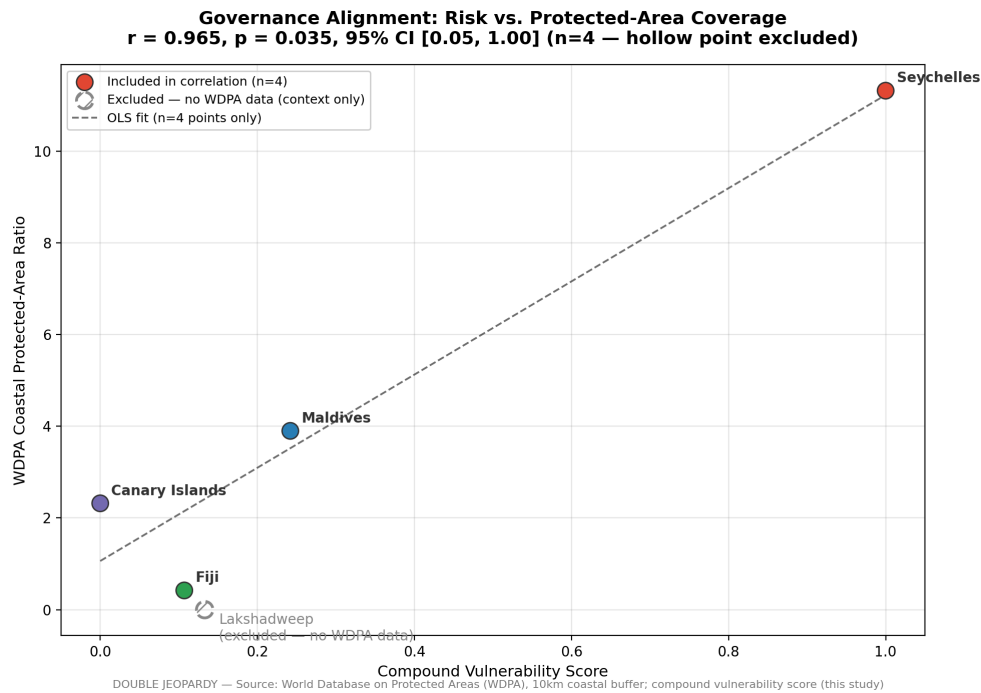


Figure 6. Compound Vulnerability Score plotted against the coastal World Database on Protected Areas (WDP) protected area ratio for each island, with a fitted ordinary least squares (OLS) reference line. Lakshadweep is shown as a hollow, hatched marker rather than a filled one, since it has no WDP measurement available; it's plotted only for visual context, and it plays no part in the regression or the correlation number below it. $r = 0.965$, $p = 0.035$, 95% confidence interval [0.05, 1.00] (n=4, Lakshadweep excluded), a positive relationship whose p value now crosses the conventional 0.05 threshold, though the 4 point sample is still too small to call this confirmatory without more data.

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Figure 7. Compound Vulnerability Score for each island as the weighting between physical exposure and coral thermal stress is swept continuously from 0% to 100%. Seychelles is the highest ranked island across the entire range, unlike an earlier version of this analysis, where Maldives overtook Seychelles beyond

approximately 76.8% physical exposure weighting. The 50/50 weighting actually used in this study isn't special or fragile in either version.

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